

Oracle SQL and Transaction Notes

Core Querying

- Use selective WHERE predicates and indexed join keys to reduce full scans.
- Always compare execution plan and row-count expectation before production deployment.
- Use bind variables for both security and parse efficiency.

DML + TCL Behavior

- INSERT UPDATE DELETE modify blocks in memory first and generate redo and undo.
- COMMIT waits for LGWR flush and makes transactions durable.
- ROLLBACK uses undo records and itself generates redo for consistency.

Interview Quick Commands

- SELECT name, open_mode, log_mode FROM v\$database;
- SELECT sql_id, executions, elapsed_time/1000000 FROM v\$sqlstats ORDER BY elapsed_time DESC FETCH FIRST 10 ROWS ONLY;
- SELECT sid, serial#, blocking_session, event FROM v\$session WHERE blocking_session IS NOT NULL;

Common Mistakes

- Running wide updates or deletes without predicate validation.
- Ignoring explain-plan regressions after stats changes.
- Committing too frequently in high-throughput workloads.